

Contents

I	Basics	1
1	Sets	1
2	Functions	2
3	Fields	3
4	The complex numbers	4
5	Equivalence relations	5
6	Modular arithmetic	5
7	Finite fields on $\mathbb{Z}/m\mathbb{Z}$	7
II	Vector spaces	8
8	Vector spaces and their properties	8
9	Subspaces of vector spaces	9
10	Linear combination and span	10

Preface

Lec 1 09-08 Tuesday

NO LATES WILL BE ACCEPTED

“work hard on your assignments”

“the test os easier than the HW because you only get an hour”

“abstraction gap”

158/240 throw it into the deep end

“very little computation in this class”

write proofs in prose more than symbols

4-5 hours outside of class

lec (tue, thu) $\rightarrow$ pset drops (fri) $\rightarrow$ tut (following thu) $\rightarrow$ pset due (sun 8pm)

I Basics

1 Sets

Lec 1 09-08 Tuesday

Definition 1.1. A set is a collection of elements. An element x is in the set S , we write $x \in S$. If the element x is not in the set S , we write $x \notin S$.

Example 1.1.1. Let $P = \{x : x \text{ is a prime number}\}$.

Definition 1.2. Two sets S and T are equal if they contain the same elements. In that case we write $S = T$.

A set can be defined either by listing the elements in the set, or by giving a characterization of these elements.

$$A = \{x : x \text{ is a positive integer dividing } 6\}$$

$$A = \{1, 2, 3, 6\} = \{2, 6, 1, 3\}$$

Definition 1.3. If S and T are sets such that all elements of S are contained in T , we say that S is a subset of T and write $S \subseteq T$. If S is a subset of T and is not equal to T , we say that S is a proper subset of T and write $S \subsetneq T$.

Definition 1.4. The empty set is the set with no elements, denoted $\emptyset$.

Definition 1.5. If S and T are sets, we have two operations

- $S \cup T = \{x : x \in S \text{ or } T \text{ or both}\}$ is called the union of S and T .
- $S \cap T = \{x : x \in S \text{ and } T\}$ is called the intersection of S and T .

Example 1.5.1. Let $\mathbb{N} = \{x \in \mathbb{Z} : x > 0\} = \{1, 2, 3, 4, \dots\}$ be the set of natural numbers, and let $-\mathbb{N} = \{x \in \mathbb{Z} : x < 0\}$ be the set of negative integers. Then $\mathbb{N} \cup -\mathbb{N} = \{x \in \mathbb{Z} : x \neq 0\}$. $\mathbb{N} \cap -\mathbb{N} = \emptyset$

Definition 1.6. Let $S_1, \dots, S_k$ sets. Then,

$$\bigcup_{i=1}^k S_i = \{x : x \in S_i \text{ for at least one } 1 \leq i \leq k\}$$

is the union of sets S_i . Likewise,

$$\bigcap_{i=1}^k S_i = \{x : x \in S_i \forall 1 \leq i \leq k\}$$

is the intersection of sets S_i .

Definition 1.7. Let S and T be sets, The **Cartesian product** is defined by $S \times T = \{(s, t) : s \in S, t \in T\}$.

2 Functions

Definition 2.1. Let S and T be non-empty sets. A function f from S to T , denoted $f : S \rightarrow T$, is a rule assigning to each $x \in S$ an unique element $f(x) \in T$.

In that case, we say that

- $f(x)$ is the **image** of x under f .
- x is the **preimage** of $f(x)$ under f .

- S is the **domain** of f .
- T is the **codomain** of f .
- the **range** of f is the set $f(S) = \{f(x) : x \in S\}$.
- if $U \subseteq S$, then the image of U is $f(U) = \{f(x) : x \in U\}$.
- if $V \subseteq T$, then the preimage of V is $f^{-1}(V) = \{x : f(x) \in V\}$.

Definition 2.2. Two functions $f, g : S \rightarrow T$ are equal if $f(x) = g(x), \forall x \in S$.

Definition 2.3. If every element in the range of f has a unique preimage, we say f is **injective**, or an **injection** (one-to-one).

Definition 2.4. If every element in the codomain is in the range, we say f is **surjective**, or a **surjection** (onto).

Definition 2.5. If f is both injective and surjective, we say f is **bijective**, or a **bijection**.

Example 2.5.1. Let $f : \mathbb{Z} \rightarrow \mathbb{N}$ be defined by $f(x) = |x| + 1$. Then f is surjective but not injective.

Definition 2.6. If $f : S \rightarrow T$ and $U \subseteq S$, then the restriction of f to U , denoted $f|_U$, is defined by $f|_U(x) = f(x)$ for all $x \in U$.

Definition 2.7. If $f : S \rightarrow T$ and $g : T \rightarrow U$ are functions, then the function which sends $x \in S$ to $g(f(x))$ is the composite $g \circ f : S \rightarrow U$, given $(g \circ f)(x) = g(f(x))$. Note even if $f, g : S \rightarrow S$, we don't necessarily have $f \circ g = g \circ f$.

Definition 2.8. A function $f : S \rightarrow T$ is **invertible** if there exists a function $g : T \rightarrow S$ such that $g \circ f(x) = x, \forall x \in S$, and $f \circ g(y) = y, \forall y \in T$. If it exists, such a g is the **inverse** of f and is denoted f^{-1} .

3 Fields

Lec 2 09-10 Thursday

Definition 3.1. A **field** F is a set with 2 operations, **note that these operations are closed**:

- $+$: $F \times F \rightarrow F$ (**sum**)
- $\cdot$: $F \times F \rightarrow F$ (**product**)

that obeys the following axioms:

- (F1) $a + b = b + a$ and $a \cdot b = b \cdot a$ (**commutativity**)
- (F2) $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (**associativity**)
- (F3) $\exists$ distinct elements $0, 1 \in F$ s.t. $0 + a = a$ and $1 \cdot a = a$ for all $a \in F$. (**existence of identities**)
- (F4) $\forall a \in F$ and $\forall b \in F, b \neq 0$, there exists $c, d \in F$ s.t. $a + c = 0$ and $b \cdot d = 1$. (**existence of inverses**)

(F5) $\forall a, b, c \in F$, then $a \cdot (b + c) = a \cdot b + a \cdot c$. (**distributivity of $\cdot$ over $+$**)

Example 3.1.1. Some fields:

- $(\mathbb{R}, +, \cdot)$
- $(\mathbb{Q}, +, \cdot)$
- $(\mathbb{Z}_2 = \{0, 1\}, +, \cdot)$

[!NOTE] memorize these axioms

[!NOTE] someone said $\mathbb{Z}/p$ lmao

[!NOTE] Can you redefine $+$ and $\cdot$ on $\mathbb{Z}$ to make it a field? Yes!

Theorem 3.2 (Cancellation Result). *Let F be a field and $a, b, c \in F$. Then: - if $a + c = b + c$, then $a = b$ - if $a \cdot c = b \cdot c$, and $c \neq 0$, then $a = b$*

Proof. Let $d \in F$ s.t. $c \cdot d = 1$.

$$(a \cdot c) \cdot d = (b \cdot c) \cdot d \quad (3.1)$$

$$a \cdot (c \cdot d) = b \cdot (c \cdot d) \quad (3.2)$$

$$a \cdot 1 = b \cdot 1 \quad (3.3)$$

$$a = b \quad (3.4)$$

A similar proof exists for the first statement. ■

Corollary 3.2.1. *The elements $0, 1 \in F$ are unique. The inverses $c, d \in F$ are also unique.*

[!NOTE] something about ring idempotent

4 The complex numbers

Definition 4.1. Let the complex numbers $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, equipped with the following operations:

- $+_C : \mathbb{C} \rightarrow \mathbb{C}$ given by $(a + bi) +_C (c + di) = (a + c) + (b + d)i$
- $\cdot_C : \mathbb{C} \rightarrow \mathbb{C}$ given by $(a + bi) \cdot_C (c + di) = (ac - bd) + (ad + bc)i$, with $i^2 = -1$ Elements of $\mathbb{C}$ are called the **complex numbers**, elements of the form $0 + bi$ are called the **imaginary numbers**.

Example 4.1.1. $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, naturally since $\mathbb{R}$ is just all the elements of the form $a + bi$.

Definition 4.2. Any complex number $z = a + bi$:

- has a **modulus** (absolute value) $|a + bi| = \sqrt{a^2 + b^2}$.
- has a **conjugate** $\bar{z} = a - bi$.

Theorem 4.3. $\mathbb{C}$ is a field.

Proof. The proof is left as an exercise to the reader. ■

5 Equivalence relations

Lec 3 09-15 Tuesday

[!quote] “I as an algebraist think they are the most important concept in algebra.”

Definition 5.1. Let S be a set. A relation R is a subset $R \subseteq S \times S$. We say that a pair $(x, y) \in S$ satisfies R if $(x, y) \in R$. We denote $(x, y) \in R$ by $x \sim y$.

Definition 5.2. A relation $\sim$ on a set S is a equivalence relation if it satisfies the following properties $\forall s, t, u \in S$:

- $s \sim s$ (**reflexivity**)
- $s \sim t \implies t \sim s$ (**symmetry**)
- $s \sim t$ and $t \sim u \implies s \sim u$ (**transitivity**)

Exercise. Consider $w \in W$, the set of all english words. What equivalence classes exist on W ?

Example 5.2.1. Some provided by people:

- $w_1 \sim w_2$, if they start with the same letter
- $w_1 \sim w_2$, if they start with the same number of characters
- $w_1 \sim w_2$, if they are the same word - **for any set you get for free a equivalence relation (equality)** - and also a trivial equivalence relation $s \sim_{tr} t, \forall s, t \in S$

Definition 5.3. If $(S, \sim)$ is a set equipped with a equivalence relation, and $t \in S$. Let **equivalence class** $C_t = \{s \in S : s \sim t\}$.

Theorem 5.4. For $(S, \sim)$, there is a subset $T \subseteq S$ that satisfies the following:

- $\bigcup_{t \in T} C_t = S$
- If $t, t' \in T$ and $t \neq t'$, then $C_t \cap C_{t'} = \emptyset$.

T is a **set of representative** of S for equivalence relation $\sim$. Note that this implies C_t partitions S , since they are all pairwise disjoint.

6 Modular arithmetic

Lec 3 09-15 Tuesday

Definition 6.1. Let $m \in \mathbb{N}$, and $a, b \in \mathbb{Z}$. We say that a is congruent to $b \bmod m$ denoted,

$$a \equiv b \bmod m \quad \text{or} \quad a \equiv b \pmod{m},$$

if $m \mid b - a$.

Example 6.1.1. The following are true:

- $11 \equiv 18 \bmod 7$
- $0 \equiv 7 \bmod 7$

Exercise. What m makes $3 \equiv 45 \pmod{m}$ true?

- $m = 1, 2, 3, 6, 7, 14, 21, 42$

Lemma 6.2. Let $m \in \mathbb{N}$, and $a, b \in \mathbb{Z}$. Then, $a \equiv b \pmod{m}$ if and only if there exists $k \in \mathbb{Z}$ s.t. $a = b + km$.

Proof. FILL IN FROM NOTES, pretty intuitive ■

Theorem 6.3 (Division with remainder). Let $m \in \mathbb{N}$. For each $a \in \mathbb{Z}$, there exists a unique $r \in \{0, 1, \dots, m-1\}$ such that $a \equiv r \pmod{m}$.

Proof. We will show that there exist unique integers k and r such that $a = km + r$ by Lemma 6.2, this give $a \equiv r \pmod{m}$.

Existence. Consider all the integer multiples of m :

$$m\mathbb{Z} = \{0, \pm m, \pm 2m, \pm 3m, \dots\}.$$

These integers occupy the entire real line by the Archimedean property of the real numbers.

![[09-15 Equivalence relations and modular arithmetic-20260915201512028.webp]]

The integer a lies somewhere on the real line, in one of these intervals. Thus, it satisfies

$$km \leq a < (k+1)m$$

for some value k . Subtract km to obtain

$$0 \leq a - km < m.$$

Let $r = a - km$. Then r satisfies the desired condition.

Uniqueness. Suppose that $a = r + km = r' + k'm$. Then

$$r - r' = (k' - k)m.$$

It follows that $r - r'$ is a multiple of m . But since $0 \leq r, r' < m$, we have $-m < r - r' < m$. The only multiple of m on this interval is 0. So $r - r' = 0 \implies r = r'$ and hence $k = k'$. ■

As a consequence, we can deduce that

$$\mathbb{Z} = \bigcup_{i=0}^{m-1} C_i$$

We call $\{0, \dots, m-1\}$ the **standard representatives** of their respective **congruence classes** C_i for congruence mod m .

Theorem 6.4 (Addition and Multiplication of Congruence Classes). If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \quad a \cdot c \equiv b \cdot d \pmod{m}$$

Proof. See pdf. ■

Definition 6.5. The ****integers modulo m (or mod m)****, denoted $\mathbb{Z}/m\mathbb{Z}$, is the set of congruence classes mod m . By the previous theorem, $\mathbb{Z}/m\mathbb{Z}$ is equipped with addition and multiplication. In practice, we will denote the congruence classes in $\mathbb{Z}/m\mathbb{Z}$ to be

$$\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{m-1}\}.$$

Remark. Let $m = 4$. Consider $\mathbb{Z}/4\mathbb{Z}$:

$$\bar{0} \cdot \bar{2} = \bar{0} \quad \bar{1} \cdot \bar{2} = \bar{2} \quad \bar{2} \cdot \bar{2} = \bar{0} \quad \bar{3} \cdot \bar{2} = \bar{2}$$

$\bar{2}$ does not have a multiplicative inverse!!!! So $\mathbb{Z}/4\mathbb{Z}$ is not a field. Note that however $\bar{3}$ does have a multiplicative inverse, it is only for (non-unity) divisors of m that lack inverses.

Theorem 6.6. $\mathbb{Z}/m\mathbb{Z}$ is a field if and only if m prime number.

Proof. Assume the field axioms except the existence of multiplicative inverses.

Assume that if m is not prime then there exists

FILL FROM PDF ■

7 Finite fields on $\mathbb{Z}/m\mathbb{Z}$

Lec 4 09-17 Thursday

FILL IN FROM NOTES

Go to tutorial!!! HW1 is due sunday at 8pm. Mathematics Support Initiative, see Quercus, math help club.

Recap. Recall that $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \dots, \overline{m-1}\}$ is the set of equivalence classes mod m , equipped with the operations $+$ and $\cdot$, given by:

1. Do the operation as usual
2. Then, take the remainder

Remark. What is a prime number?

Definition 7.1. Let $p \in \mathbb{N}$, $p > 1$, then p is prime if $\forall x, y \in \mathbb{Z}$,

$$p \nmid x \text{ and } p \nmid y \implies p \nmid xy$$

asfasf

Remark. What is a prime number?

Check this as an exercise! This is also called Euclid's lemma.

Theorem 7.2. Let $m \in \mathbb{N}$. If m is prime, then $\mathbb{Z}/m\mathbb{Z}$ is a field. *blahblahblahblahblahblah
blah blah blah blah*

Remark. What is a prime number?

Proof. First, check the field axioms, with (F4) holding for addition. ■

II Vector spaces

8 Vector spaces and their properties

Lec 5 09-22 Tuesday

Review of vectors from HS, check notes pdf.

Definition 8.1. A vector space over a field F is a set V equipped with two (closed) operations,

- $+: V \times V \rightarrow V$ (addition)
 $(x, y) \mapsto x + y$
- $\cdot: F \times V \rightarrow V$ (scalar multiplication)
 $(a, x) \mapsto ax$

and also satisfies the eight following axioms $\forall x, y, z \in V$ and $\forall a, b \in F$.

- (VS1) $x + y = y + x$ (+ is commutative)
- (VS2) $(x + y) + z = x + (y + z)$ (+ is associative)
- (VS3) $\exists 0 \in V$ s.t. $x + 0 = x$ (+ has an identity)
- (VS4) $\forall x \exists y$ s.t. $x + y = 0$ (+ has inverses)
- (VS5) $1x = x$
- (VS6) $a(bx) = (ab)x$
- (VS7) $a(x + y) = ax + ay$
- (VS8) $(a + b)x = ax + bx$

Remark. Note that (VS1-4) is equivalent to saying that $(V, +)$ forms an abelian group, and that F is only relevant for the latter four axioms. Also, note that $0 \in V$ and $0 \in F$ is not necessarily exactly the same compounds.

Remark. A vector space *need not* have multiplication defined on vectors!

Example 8.1.1. Some vector spaces include:

- Every field is a vector space over itself. (pretty trivial)
- $F^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in F\}$ for any field F is a vector space over F .
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$, and also a vector space over $\mathbb{Q}$!

- Let $k \in \mathbb{Z}_{>0}$. $P_k(F) = \{a_0 + a_1x + \cdots + a_kx^k \mid a, b \in F\}$, the set of polynomials with degree k and coefficients from F , is a vector space over F .
- $P(f) = \bigcup_{k \in \mathbb{Z}_{>0}} P_k(F)$ is a vector space of all polynomials over F .
- Let S be a nonempty set, and a F be a field. $\mathcal{F}(S, F)$ is the set of all functions from $S \rightarrow F$. $\mathcal{F}$ is a vector space over F , with addition and scalar multiplication defined as follows, for $f, g \in \mathcal{F}$, $s \in S$, $a \in F$.

$$(f + g)(s) = f(s) + g(s)$$

$$(af)(s) = a \cdot f(s)$$

Theorem 8.2 (Simple properties of vector spaces). *Let V be a vector space over F , $v, x, y, z \in V$, and $a \in F$. Then*

- (i) $0 \cdot v = 0$.
- (ii) *The 0 vector is unique.*
- (iii) $a \cdot 0 = 0$.
- (iv) $x + z = y + z \implies x = y$.
- (v) $-x$ is defined as $x + (-x) = 0$. *It is unique.*
- (vi) *If $a \in F$, then $(-a)b = a(-v) = -(av)$.*

Proof. pretty simple, do as exercise ■

Definition 8.3. matrices fill in from notes

9 Subspaces of vector spaces

Definition 9.1. Let V be a vector space over F . $W \subseteq V$ is a **subspace** of V if it is a vector space equipped with the restriction of operations of V .

Theorem 9.2 (VERY IMPORTANT AND USEFUL). *Let V be a vector space over F . Then $W \subseteq V$ is a subspace if and only if it satisfies:*

- (i) $W \neq \emptyset$
- (ii) W is closed under addition
- (iii) W is closed under scalar multiplication

Proof. SEE PDF ■

Example 9.2.1. Consider the vector space $\mathbb{R}^3 = \{(a, b, c) \mid a, b, c \in \mathbb{R}\}$ over $\mathbb{R}$.

- $\mathbb{R}^3$ is a subspace of itself. (trivial)
- $W_1 = \{(a, b, c) \mid c = 0\}$ is a subspace. ($\cong \mathbb{R}^2$)

- $W_2 = \{(a, b, c) \mid 71a - 22b + 7c = 0\}$ is a subspace. ($\cong \mathbb{R}$)
- $W_3 = \{(a, b, c) \mid a + b = -c^2\}$ is NOT a subspace. ($v = (-2, -2, -2) \in W_3$ but $2v = (-4, -4, -4) \notin W_3$.)

Theorem 9.3. Let V be a vector space and W_1 and W_2 be subspaces of V . Then, $W_1 \cap W_2$ is also a subspace.

Proof. We will prove the conditions in Theorem 9.2

- (i) $0 \in W_1$ and $0 \in W_2$, so $0 \in W_1 \cap W_2$.
- (ii) $x, y \in W_1 \cap W_2 \implies x + y \in W_1$ and $x + y \in W_2$ because addition is closed in subspaces, so $x + y \in W_1 \cap W_2$.
- (iii) $x \in W_1 \cap W_2 \implies cx \in W_1$ and $cx \in W_2$ because scalar multiplication is closed in subspaces, so $cx \in W_1 \cap W_2$. ■

10 Linear combination and span

Lec 6 09-24 Thursday

Definition 10.1. Let V be a vector space, and S be nonempty subset of V . We say $v \in V$ is a **linear combination** of $s \in S$ if there exists $u_1, u_2, \dots, u_n \in S$ and $a_1, a_2, \dots, a_n \in F$ s.t.

$$v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$$

We call a_i **coefficients**.

Example 10.1.1. 0 is always a linear combination of any set of vectors.

Example 10.1.2. Consider the vector space of polynomials $P(\mathbb{Q})$ over $\mathbb{Q}$. $3x^2 + 1$ is a linear combination of x^2 and 1, or even $-7x^2 + 1$ and $4x^2 + 2$. (The reader may check the latter as an exercise.)

Remark. Many applications of linear algebra have to do with representations of linear combinations, see urban planning example in pdf.

Exercise. Draw the linear combinations of:

- (i) $(1, 1)$ in $\mathbb{R}^2$ INSERT IMAGE
- (ii) $(1, 0, 1)$, $(2, 0, 2)$, and $(0, 0, 1)$ in $\mathbb{R}^3$ INSERT IMAGE

Definition 10.2. Let S be a nonempty subset of vector space V . The **span** of S is the set of linear combinations of S , denoted $\text{span}(S)$. By definition, $\text{span}(\emptyset) = \{0\}$.

Theorem 10.3. Let S be a subset of a vector space V . $\text{span}(S) = W$ is a subspace of V . Also, $S \subseteq W$. Moreover, any subspace of V that contains S , also contains $\text{span}(S)$

Proof. Using Theorem 9.2 we show that $\text{span}(S)$ is a subspace by the following:

- (i) $0 \in \text{span}(S)$ (see Example [10.1.1](#)) even if S is empty, since by definition $\text{span}(\emptyset) = \{0\}$. So, $\text{span}(S)$ is not empty.
- (ii) $\text{span}(S)$ is closed under addition. too lazy to write this out, its intuitive.
- (iii) $\text{span}(S)$ is closed under scalar multiplication. also too lazy lmao, same logic, just write out the definitions.

For every $s \in S$, s can be written as a linear combination of itself by $1 \cdot s = s$, so it is in $\text{span}(S) = W$, so $S \subseteq W$.

Finally, we show that every subspace of V that contains S also contains $\text{span}(S)$. By definition, subspaces are closed under addition and scalar multiplication, which are exactly the operations that make up linear combinations. So, $\text{span}(S)$ must also be in that subspace, as it is exactly the linear combinations of S . ■